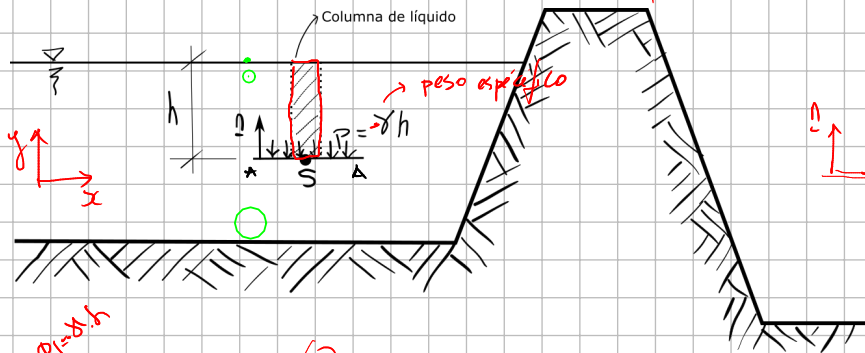
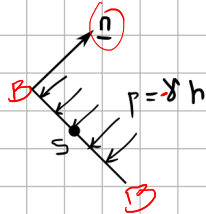
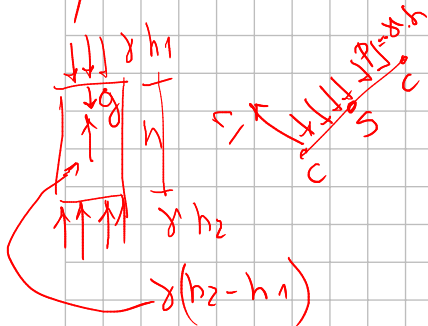
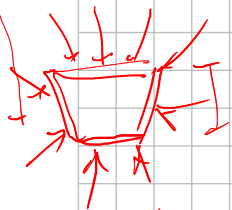


# Hidroestática

Fluido incompresible



$$T_y = -\gamma \cdot h \cdot dA \equiv \gamma_y$$



$$P = -\gamma h$$

$$\underline{\underline{T}} = \begin{bmatrix} P & 0 & 0 \\ 0 & P & 0 \\ 0 & 0 & P \end{bmatrix}$$

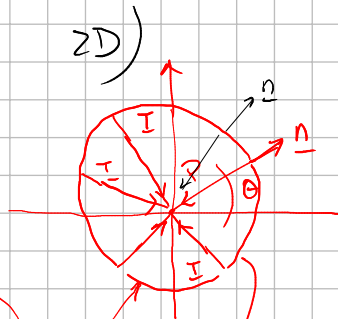
$$\underline{\underline{T}} = \underline{\underline{T}} \cdot \underline{\underline{n}} = \begin{bmatrix} P & 0 & 0 \\ 0 & P & 0 \\ 0 & 0 & P \end{bmatrix} \cdot \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

$$\underline{\underline{T}} = P \cdot \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

$$= \begin{bmatrix} P \cdot n_x \\ P \cdot n_y \\ P \cdot n_z \end{bmatrix} = P \cdot \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}$$

$$\|\underline{\underline{T}}\| = |P| \|\underline{\underline{n}}\|$$

$$\|\underline{\underline{T}}\| (n_x, n_y, n_z) = |P|$$



una circunferencia

3D) Esfera

$$\underline{\underline{T}} = \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{xy} & \tau_{yy} \end{bmatrix}$$

